

Homework (Carolina Reaper)

Q1. Factor the expression: $x^2 + 5x + 4x + 20$

- (A) $x(x + 5) + 4(x + 5)$
(B) $x(x + 4) + 5(x + 4)$
(C) $x(x + 1) + 4(x + 20)$
(D) $x(x + 20) + 5(x + 4)$
(E) $x(x + 4) + 1(x + 20)$

Q2. A painter is creating a canvas with a rhythm pattern of $3x + 2x + 12 + 8$. Factor the expression to find the pattern:

- (A) $5x + 20$
(B) $x(5x + 20)$
(C) $5(x + 4)$
(D) $x(3 + 2) + 4(3 + 2)$
(E) $5x + 4(3 + 2)$

Q3. Factor the expression: $6x^2 + 18x + 2x + 6$

- (A) $6x(x + 3) + 2(x + 3)$
(B) $3x(2x + 6) + 1(2x + 6)$
(C) $6(x + 3) + 2(x + 3)$
(D) $2(3x^2 + 9x + x + 3)$
(E) $3(2x^2 + 6x +$

$\frac{1}{2}23x + 2)$

Q4. A DJ is mixing frequencies at $2x + 6x + 4 + 24$. Factor the expression to find the beat: $2x + 6x + 4 + 24$

- (A) $8x + 28$
(B) $2(4x + 12)$
(C) $8(x +$

$\frac{1}{2}72)$

(C) $4(2x + 6) + 4$

(D) $2(4x + 12)$

Q5. Factor the expression: $9x^2 + 12x + 3x + 4$

- (A) $3x(3x + 4) + 1(3x + 4)$
(B) $9(x^2 +$

$\frac{1}{2}43x) + 4$

(B) $3(3x^2 + 4x + x +$
 $\frac{1}{2}43)$

(B) $3x(3x + 1) + 4(3x + 1)$

(C) $3x(3x + 4) + 4(1)$

Q6. A music producer has a rhythm pattern of $x^2 + 4x + 5x + 20$. Factor the expression: $x^2 + 4x + 5x + 20$

- (A) $x(x + 4) + 5(x + 4)$
(B) $x(x + 5) + 4(x + 5)$
(C) $x(x + 9) + 4$
(D) $x^2 + 9x + 20$
(E) $x(x + 4) + 5(x + 4)$

Q7. Factor the expression: $8x^2 + 10x + 2x + 5$

(A) $2(4x^2 + 5x + x +$

$\frac{1}{2}52)$

(A) $2(4x^2 + 5x) + 1(2x + 5)$

(B) $5(1x^2 + 2x) + 2(1x +$
 $\frac{1}{2}52)$

(B) $2(4x^2 + 5x) + 2x + 5$

(C) $2(4x^2 + 5x + x +$
 $\frac{1}{2}52)$

Q8. A sculptor has a frequency of $4x^2 + 12x + 3x + 9$. Factor the expression: $4x^2 + 12x + 3x + 9$

(A) $3($

$\frac{1}{2}43 x^2 + 4x + x + 3)$

(B) $4(x^2 + 3x) + 3(x + 3)$

(C) $3($

$\frac{1}{2}43 x^2 +$
 $\frac{1}{2}123 x + x + 3)$

(C) $3x($
 $\frac{1}{2}43 x + 4) + 3(x + 3)$

(C) $3(x + 3)($

$\frac{1}{2}43x + 1)$

Open Ended Questions (FRQ)

Q9. Factor the expression $x^2 + 7x + 2x + 14$ and verify by multiplication. Provide all steps.

Q10. A photographer is working with a canvas size of $x^2 + 9x + 4x + 36$. Factor the expression and verify the result using multiplication. Show all work.

Chef's Tips

- Chef's Tips: Ensure students show all work for factoring and verifying the results
- Remind students to use the distributive property for verification
- Encourage students to use rhythm patterns and canvas sizes to visualize the factoring process